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USER MANUAL

Bell Aid: user manual

A light that flashes when your doorbell rings

PROJECT	VERSION	DATE
Bell Aid	1.3	12 November 2026

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What it does

The **sender** stands near your doorbell and listens for its sound. When it hears the bell, every paired **receiver** in your home flashes its light ring. No Wi-Fi, no app, no cloud: the sound is learned and recognised on the sender itself. Sender and receiver look the same; one sender works with up to 5 receivers. One light of the ring is the status light D1.

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What you need

- The assembled sender and one or more receivers, each in its stand (see the build guide)
- Per unit: one USB-C cable and one USB wall plug, **5 V, at least 1.5 A**

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The button

The button acts when you **release** it; a short press acts about 0.4 s later, because a second press may follow. While you hold it, the status light D1 turns **yellow** at 1.5 s and **magenta** at 5 s, so you know when to let go.

PRESS	SENDER	RECEIVER
Short (< 1.5 s)	Send a test ring (the sender also flashes for 3 s); while flashing: stop the flash on this unit only	Mute on/off; while flashing: stop the flash
Long (release on yellow)	Start training	Test its own light (3 s)
Very long (release on magenta)	Pairing (30 s)	Pairing (60 s)
3 short presses in a row (each pause up to 0.4 s)	Green light (heartbeat) on/off	Green light (heartbeat) on/off

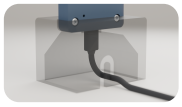
Triple press: D1 shows green twice (on) or one slow green fade-out (off). Each unit keeps its own setting, also after unplugging. Two presses, or four or more, do nothing.

Tip: hold the stand with one hand while you press the button; it tips over easily.

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Set up

1. Plug in. Lift the unit out of its stand and plug the cable in while you hold the unit in your hand. Then lower it into the stand, hook the cable into the J slot in the back wall and lead it down and to the rear. To unplug: unhook the cable, lift the unit out, then pull the plug.



Back view, stand see-through: cable in the slot. Rendering of the CAD model, not a photograph.

2. Choose the sender. Every unit starts as a receiver. To make one unit the sender: with the unit in your hand (out of its stand, as in step 1), hold its button while you plug in the cable and keep holding for 5 s. D1 turns yellow, then blinks green 3 times, and the unit restarts as the sender. The same gesture switches it back.

3. Pair each receiver.

1. Plug in the receiver. A new receiver waits for pairing for 60 s (D1 blinks fast blue). Otherwise hold its button very long (magenta).
2. Hold the sender's button very long and release on magenta. D1 blinks magenta for 30 s.
3. The receiver blinks green 3 times: paired. Repeat for each receiver.

4. Place the sender near the doorbell chime, where it will stay. Keep the small microphone hole on the back free.

5. Teach it your doorbell. Keep the room quiet.

1. Hold the sender's button and release on yellow. D1 pulses blue: waiting.
2. Ring the doorbell and let it play to the end. D1 is cyan while it listens, then flashes green.
3. Wait about 3 s and ring again; 3 rings in total.
4. D1 blinks green 3 times: saved. Red 3 times: no ring heard, start again.
5. **Check:** wait a few seconds and ring once more. The sender's whole ring and the receivers must flash. If they do not, train again.

> NOTE
> With a very short bell (a single "ding"), the click of the bell push can be learned with it. If the check keeps failing, have someone else press the push with the door closed while you stay at the sender, and train again.

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Everyday use

When the bell rings, the sender and each receiver flash the whole ring for 15 s (test: 3 s): white, then six amber lights chasing clockwise. A short press stops the flash on that unit only. The flash never plays during training or pairing. A muted receiver shows only test rings until you unmute or unplug it.

STATUS LIGHT D1	SENDER	RECEIVER
Green blip every 3 s (heartbeat)	Ready	Ready (paired)
Orange	No doorbell learned yet	Not paired
Whole ring flashes	Bell heard or test sent (stop: short press)	Bell heard (stop: short press)
Dim red, steady		Muted
Magenta / fast blue blink	Pairing	Waiting for pairing
Blue pulse, cyan, green flash	Training: waiting, listening, ring captured	
Green twice / one slow green fade-out	Triple press: heartbeat switched on / off	Triple press: heartbeat switched on / off
Green 3x / red 3x	Training saved / failed	Paired

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Quick fixes

PROBLEM	WHAT TO DO
No light at all	Check the wall plug (5 V, 1.5 A) and try another cable.
Orange light	Receiver: pair it (4, step 3). Sender: train it (4, step 5).
Test not received	Pair the receiver again; test it closer to the sender (walls cut range).
Bell not detected	Train again in place; keep the microphone hole free. Short bell: note in 4.
Green blink bothers me at night	Press the button 3 times in a row on each unit; repeat to switch it on again.
False alarms	Train again with 3 rings in a quiet room.

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Updating the software

Settings and pairing stay. If the light no longer reacts to your bell, teach the bell sound again (section 4, step 5). You need **Chrome or Edge** on a computer and a data USB cable (some charge-only cables do not work).

1. Plug in **one** device. Open https://makehare.github.io/bell-aid/update/, click **Update**.
2. Pick **USB JTAG/serial debug unit**, Connect, then **Install Bell Aid**. Leave **Erase device** unticked (if ticked, role, pairing and the trained doorbell sound are lost), click **Next**, then **Install**. Wait for **Installation complete!** (about 2 minutes), click **Next**. The window is in English.
3. Unplug. Repeat for each device, one at a time.

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Safety, radio data and disposal

Intended use: only as a visual doorbell signal in the home. **DIY open-source project:** you build and use it at your own risk; it is not a certified product.

- Indoors and in dry rooms only. Keep away from water, heat and direct sun.
- **Not a smoke or fire alarm** and no substitute; do not use where a missed signal is dangerous.
- Not a toy; keep away from small children. Unplug it before you open the case or if it is damaged.

TECHNICAL DATA

Frequency band	2.4 GHz (2400–2483.5 MHz)
Max. RF power	20 dBm (100 mW) conducted, ESP32-S3 setting
Power supply	5 V DC via USB-C, at least 1.5 A recommended
Help and issues	Makehare, request@makehare.com
Open-source files	https://github.com/Makehare/bell-aid



Disposal: the crossed-out wheeled bin means: do not put it in household waste; take it to an electronics collection point (e.g. your recycling centre).

Before you hand it in: the unit stores no personal data, only its role, pairing data and the learned doorbell sound. To erase these, flash the complete firmware image once more (build guide, section 6.2).